THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 10 — MATHS · MEDIUM LEVEL UNIT TEST SET 3

Chapters 5–7: **Arithmetic Progressions, Triangles, Coordinate Geometry** | RBSE Pattern | Session 2026–27

Time: 2 Hours Name: _____ Roll No.: _____ **Maximum Marks: 50**

General Instructions:

1. All questions are compulsory.
2. Q. 1–15 are Objective Type (1 mark each).
3. Q. 16–20 are Very Short Answer Type (2 marks each).
4. Q. 21–25 are Short Answer Type (3 marks each).
5. Q. 26–27 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 24–25 and Q. 26–27.
7. Write the question number before each answer.
8. Medium level: multi-step questions.

SECTION — A (Objective Type Questions) 15 × 1 = 15

Choose the correct option and write it in your answer book.

1. The middle term of an AP with 11 terms, first term 8 and last term 48, is: 1

(a) 30 **(b)** 29 **(c)** 27 **(d)** 28

2. The diagonals of a rhombus are 10 and 24. Its side squared is: 1

(a) 170 **(b)** 169 **(c)** 168 **(d)** 171

3. The ratio $k:1$ in which the line $x=4$ divides the join of $(1,0)$ and $(10,0)$, with $a<4<c$, has $k=$ 1

(a) $-\frac{1}{2}$ **(b)** $\frac{3}{2}$ **(c)** $\frac{5}{2}$ **(d)** $\frac{1}{2}$

4. If $a_{12}=32$ and $d=2$, then $a=$ 1

(a) 10 **(b)** 9 **(c)** 12 **(d)** 11

5. Two poles of heights 20 m and 14 m are 8 m apart. The distance between their tops (m) is: 1

(a) 10 **(b)** 12 **(c)** 11 **(d)** 9

6. If $(0,1)$, $(2,5)$ and $(4,k)$ are collinear, then $k=$ 1

(a) 9 **(b)** 11 **(c)** 8 **(d)** 10

7. The 15th term of the AP whose 1st term is 11 and 6th term is 21 is: 1

(a) 40 **(b)** 41 **(c)** 38 **(d)** 39

8. If $DE \parallel BC$ in $\triangle ABC$, $AB=8$ cm, $AD=2$ cm, $AC=16$ cm, then EC (cm) is: 1

(a) 11 **(b)** 14 **(c)** 12 **(d)** 13

9. The perimeter of the triangle with vertices $(0,0)$, $(9,0)$, $(0,12)$ is: 1

(a) 38 **(b)** 37 **(c)** 35 **(d)** 36

10. The number of terms in the AP 2, 8, \dots, 98 is: 1

(a) 17 **(b)** 16 **(c)** 18 **(d)** 19

11. The areas of two similar triangles are 16 cm² and 81 cm². The ratio of their perimeters is 4: 1

(a) 10 **(b)** 11 **(c)** 8 **(d)** 9

12. The point which divides $(12,0)$ and $(0,6)$ in the ratio 1:1 has coordinates summing to: 1

(a) 8 **(b)** 11 **(c)** 10 **(d)** 9

13. The sum of all two-digit multiples of 5 is: 1

(a) 944 **(b)** 947 **(c)** 945 **(d)** 946

14. The altitude to the hypotenuse of a right triangle divides it into 2 cm and 18 cm. The altitude (cm) is: 1

(a) 6 **(b)** 5 **(c)** 8 **(d)** 7

15. The centroid of the triangle with vertices $(2, 5)$, $(8, 1)$, $(5, 6)$ has x-coordinate: 1

(a) 4 **(b)** 6 **(c)** 5 **(d)** 7

SECTION — B (Very Short Answer Type) 5 × 2 = 10

Answer in about 20–30 words each.

16. In $\triangle ABC$, D and E lie on AB and AC with $DE \parallel BC$. If $AB = 10$ cm, $AD = 4$ cm and $AC = 20$ cm, find AE and EC. 2

17. Find the values of y for which the distance between $P(9,\ y)$ and $Q(3,\ -1)$ is 10 units. 2

18. Find the sum of the first 26 terms of the AP $-12, -8, -4, 0, \ldots$ 2

19. Two similar triangles have perimeters 16 cm and 48 cm. If a side of the smaller is 5 cm, find the corresponding side of the larger and the ratio of their areas. 2

20. Three vertices of a parallelogram $ABCD$ taken in order are $A(0,\ 0)$, $B(4,\ 1)$ and $C(6,\ 5)$. Find the fourth vertex D. 2

SECTION — C (Short Answer Type) 5 × 3 = 15

Answer in about 40–60 words each.

21. Find the coordinates of the point which divides the join of $(1,\ -4)$ and $(7,\ 8)$ internally in the ratio $3 : 1$. 3

22. The sum of the first 12 terms of an AP is 360 and the first term is 8. Find d and the 12th term. 3

23. In a right triangle, the altitude to the hypotenuse divides it into segments of 2 cm and 8 cm. Find the altitude. 3

24. Find the area of the triangle with vertices $(-4,\ -2)$, $(2,\ 4)$ and $(6,\ -4)$. 3

OR

In an AP, the 4th term is 3 and the 9th term is -12. Find the AP and its 14th term. 3

25. In $\triangle ABC$, D and E lie on AB and AC with $DE \parallel BC$. If $AB = 9$ cm, $AD = 6$ cm and $AC = 18$ cm, find AE and EC. 3

OR

Show that the points $(-2,\ 0)$, $(2,\ 0)$ and $(0,\ 4)$ are the vertices of an isosceles triangle. 3

SECTION — D (Long Answer Type) 2 × 5 = 10

Answer in about 100–150 words each.

26. In what ratio does the x-axis divide the segment joining $(1,\ -8)$ and $(4,\ 7)$? Find the point of division. 5

OR

A ladder has 13 rungs whose lengths decrease uniformly from 60 cm at the bottom to 30 cm at the top. Find the length of the wood needed for the rungs and the length of the middle rung. 5

27. The diagonals of a rhombus are 10 cm and 24 cm. Find its side. 5

OR

Find the point on the x-axis which is equidistant from $(-3,\ 2)$ and $(5,\ 6)$. 5

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END OF QUESTION PAPER **Total: 50 Marks**

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